

# Exercise: FAST Interaction Detection with Prefix Sums

## Exercise

The table lists  $n = 9$  samples with two numerical features and a target:

| Idx | $x_1$ | $x_2$ | $y$ |
|-----|-------|-------|-----|
| 0   | 1.07  | 1.11  | 2   |
| 1   | 1.86  | 1.05  | 3   |
| 2   | 3.18  | 1.07  | 4   |
| 3   | 0.93  | 2.16  | 5   |
| 4   | 2.12  | 2.08  | 6   |
| 5   | 2.85  | 2.14  | 7   |
| 6   | 1.18  | 3.06  | 8   |
| 7   | 2.03  | 2.92  | 9   |
| 8   | 3.09  | 3.17  | 10  |

## Tasks

1. Discretize each feature into three equal-width bins  $[0, 1.5) \rightarrow 0$ ,  $[1.5, 2.5) \rightarrow 1$ ,  $[2.5, 3.5) \rightarrow 2$ .
2. Build two  $3 \times 3$  matrices

$$S(i, j) = \sum_{(x_1^b, x_2^b) = (i, j)} y, \quad N(i, j) = \sum_{(x_1^b, x_2^b) = (i, j)} 1.$$

3. Form 2-D prefix sums  $S^{\text{pref}}, N^{\text{pref}}$ .
4. Via inclusion-exclusion obtain totals for the rectangle  $x_1^b \in \{1, 2\}$ ,  $x_2^b \in \{1, 2\}$ :  $S_r$ ,  $N_r$ .
5. Compute the mean in this region:  $\hat{y}_r = S_r / N_r$ .
6. Let  $S_n = \sum_{i=1}^n y^{(i)}$ ,  $N_n = n$ . Show that the *RSS reduction*

$$\Delta \text{RSS} = \frac{S_r^2}{N_r} + \frac{(S_n - S_r)^2}{N_n - N_r} - \frac{S_n^2}{N_n}$$

needs only first-order sums (no  $y^2$ ).

## Solution

### 1. Binning the data

| Idx | $x_1$ | $x_2$ | $y$ | $(x_1^b, x_2^b)$ |
|-----|-------|-------|-----|------------------|
| 0   | 1.07  | 1.11  | 2   | (0,0)            |
| 1   | 1.86  | 1.05  | 3   | (1,0)            |
| 2   | 3.18  | 1.07  | 4   | (2,0)            |
| 3   | 0.93  | 2.16  | 5   | (0,1)            |
| 4   | 2.12  | 2.08  | 6   | (1,1)            |
| 5   | 2.85  | 2.14  | 7   | (2,1)            |
| 6   | 1.18  | 3.06  | 8   | (0,2)            |
| 7   | 2.03  | 2.92  | 9   | (1,2)            |
| 8   | 3.09  | 3.17  | 10  | (2,2)            |

## 2. Aggregate matrices $S$ and $N$

$$S = \begin{bmatrix} 2 & 5 & 8 \\ 3 & 6 & 9 \\ 4 & 7 & 10 \end{bmatrix}, \quad N = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

## 3. 2-D prefix sums

$$S^{\text{pref}} = \begin{bmatrix} 2 & 7 & 15 \\ 5 & 16 & 33 \\ 9 & 27 & 54 \end{bmatrix}, \quad N^{\text{pref}} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

## 4. Rectangle (1:2, 1:2)

Let the binned grid have indices  $0, \dots, B-1$  on each axis and define an axis-aligned rectangle

$$R = [i_{\min} : i_{\max}] \times [j_{\min} : j_{\max}], \quad 0 \leq i_{\min} \leq i_{\max} < B, \quad 0 \leq j_{\min} \leq j_{\max} < B.$$

With 2-D prefix sums

$$S^{\text{pref}}(i, j) = \sum_{u \leq i, v \leq j} S(u, v), \quad N^{\text{pref}}(i, j) = \sum_{u \leq i, v \leq j} N(u, v),$$

the totals in  $R$  are obtained via inclusion-exclusion:

$$\begin{aligned} S_R &= S_{i_{\max}, j_{\max}}^{\text{pref}} - S_{i_{\min}-1, j_{\max}}^{\text{pref}} - S_{i_{\max}, j_{\min}-1}^{\text{pref}} + S_{i_{\min}-1, j_{\min}-1}^{\text{pref}}, \\ N_R &= N_{i_{\max}, j_{\max}}^{\text{pref}} - N_{i_{\min}-1, j_{\max}}^{\text{pref}} - N_{i_{\max}, j_{\min}-1}^{\text{pref}} + N_{i_{\min}-1, j_{\min}-1}^{\text{pref}}. \end{aligned} \tag{1}$$

(All terms with index  $-1$  are interpreted as zero.)

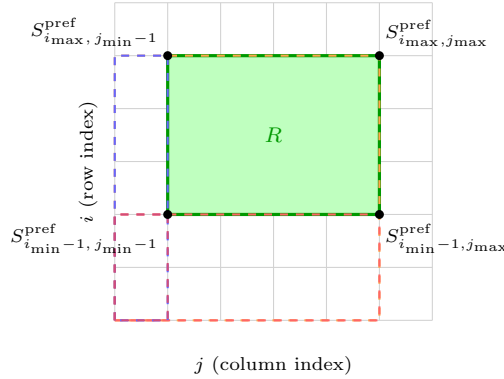


Figure 1: Inclusion-exclusion on a 2-D prefix grid. The green area is the query rectangle  $R$ . Dashed rectangles show the four prefix sums employed in Eq. (1): orange (+), red and blue (-), purple (+).

Using the same four look-ups one can obtain the subtotals of the *other three* regions that partition the orange prefix rectangle:

$$\begin{aligned} S_{\text{top}} &= S_{i_{\min}-1, j_{\max}}^{\text{pref}} - S_{i_{\min}-1, j_{\min}-1}^{\text{pref}}, \\ S_{\text{left}} &= S_{i_{\max}, j_{\min}-1}^{\text{pref}} - S_{i_{\min}-1, j_{\min}-1}^{\text{pref}}, \\ S_{\text{topleft}} &= S_{i_{\min}-1, j_{\min}-1}^{\text{pref}}. \end{aligned}$$

(Replace  $S^{\text{pref}}$  by  $N^{\text{pref}}$  to get the corresponding counts.)

**RSS drop using only first-order sums.** Let  $S_n = \sum_{i=1}^n y^{(i)}$  and  $N_n = n$ . Define  $S_C = S_n - S_R$  and  $N_C = N_n - N_R$ . Then the reduction in residual sum of squares when isolating rectangle  $R$  is

$$\Delta \text{RSS}(R) = \frac{S_R^2}{N_R} + \frac{S_C^2}{N_C} - \frac{S_n^2}{N_n}, \tag{2}$$

which involves *only* the first-order target sums  $S_\bullet$  and counts  $N_\bullet$ ; all  $\sum y^2$  terms cancel. This identity is the core of the FAST interaction-search algorithm and the 1-D prefix-sum split optimisation presented in the lecture.

Using inclusion-exclusion:

$$\begin{aligned} S_r &= S_{2,2}^{\text{pref}} - S_{0,2}^{\text{pref}} - S_{2,0}^{\text{pref}} + S_{0,0}^{\text{pref}} = 54 - 15 - 9 + 2 = 32, \\ N_r &= N_{2,2}^{\text{pref}} - N_{0,2}^{\text{pref}} - N_{2,0}^{\text{pref}} + N_{0,0}^{\text{pref}} = 9 - 3 - 3 + 1 = 4. \end{aligned}$$

## 5. Mean prediction

$$\hat{y}_r = \frac{S_r}{N_r} = \frac{32}{4} = 8.$$

## 6. RSS reduction with first-order sums only

Total sums:  $S_n = 54$ ,  $N_n = 9$ . Complement:  $S_c = S_n - S_r = 22$ ,  $N_c = N_n - N_r = 5$ .

$$\Delta\text{RSS} = \frac{S_r^2}{N_r} + \frac{S_c^2}{N_c} - \frac{S_n^2}{N_n} = \frac{32^2}{4} + \frac{22^2}{5} - \frac{54^2}{9} = 28.8$$

*Observation:* The formula contains only the sums  $S$  and counts  $N$ ; all  $\sum y^2$  terms cancel. This is exactly the trick exploited by FAST and the prefix-sum split algorithm in the lecture.